

Unlocking Behavioral Intelligence in Airline Loyalty Programs

Technical Report

Churn Prediction · Customer Segmentation · Smart Retention

EXECUTIVE SUMMARY

This report delivers a data-driven retention strategy for a mid-sized Canadian airline's loyalty program, covering 16,737 members (2012–2018). The mandate: move beyond points-and-rewards thinking to build a proactive system that identifies at-risk members before they leave, segments them by value, and prescribes specific interventions the marketing team can deploy tomorrow.

KEY FINDINGS AT A GLANCE

- ▶ **Churn Rate:** 7.8% (1,310 of 16,737 members flagged as churned)
- ▶ **Churn composition:** 1,147 formal cancellations · 130 silent disengages ,33 late-stage vanishers
- ▶ **Model:** Random Forest, 5-fold OOF AUC = 0.840 ± 0.003 no data leakage confirmed
- ▶ **1,602 High-Risk members identified for immediate action**
- ▶ **Premium Loyalists:** 6.6% of members, ~4.4× average CLV (£27,729 vs £8,000 mean)
- ▶ **Dormant Members:** highest churn rate (11.5%), largest High-Risk count (628)
- ▶ **Output:** 12-cell retention matrix every action specifies who, when, what, channel, and why

1. PROBLEM FRAMING & DATA CLEANING

1.1 Churn Definition :Three Signals

The dataset contains formal cancellation records, but cancellation is not the only form of churn. We defined churn through three mutually reinforcing signals, using 2018 data only for label construction never as a feature (leakage prevention).

Signal	Definition	Count
Signal 1 Formal Churn	Cancelled in 2017 or 2018. Ground truth. Pre-2017 cancellations excluded they left before our prediction window.	1,147 members (87.6%)
Signal 2 Silent Disengagement	Enrolled pre-2017 + zero flights in 2017 + CLV above median (~£5,780) + no formal cancel. CLV filter removes never-active 2012 sign-ups.	130 members (9.9%)
Signal 3 Late-Stage Risk	Active in 2017 but completely silent in 2018 with no cancellation an engaged member who vanished.	33 members (2.5%)

Why 7.8% is correct: The program grew in 2018 (12,439 → 14,614 active members). A higher rate would flag never-engaged members; a lower rate would miss genuine silent losses. 7.8% is real, contained, and concentrated in formal cancellations with a meaningful silent layer.

1.2 Data Cleaning Decisions

Every decision preserved data rather than discarding it where possible:

Issue	Decision & Rationale
Negative salaries	Replaced with NaN. Then imputed via group median by Education × Marital Status salary correlates with both. 4,258 nulls resolved; 0 remaining.
Activity duplicates	Type A (exact row copies): dropped first. Type B (split bookings same month): aggregated by summing metrics within Loyalty Number × Year × Month.
Impossible cancellations	Cancellation month = enrollment month is operationally impossible. Nullified cancellation fields; conservative keeps the member as active rather than discarding the record.
Negative activity values	Clipped to zero. Negative flight counts/distances are not physically possible; clipping preserves the member record.
CLV outliers (168 members)	Flagged with binary clv_outlier at 99th percentile (£35,929). Kept in dataset; flag lets the model treat them distinctly without distorting distributions.
Ordinal encoding	Loyalty Card: Star=1, Nova=2, Aurora=3. Education: High School=1 → Doctor=5. From data dictionary's explicit rankings.
Calendar integration	Daily calendar (2,557 rows) distilled to monthly lookup; merged into activity to inject Quarter + Season. All 24 month-year combos confirmed present zero orphan joins.

2. FEATURE ENGINEERING & MODELLING

2.1 Feature Categories

All 25+ features are derived exclusively from 2017 and earlier data. Design logic: behavioral signals over demographic proxies.

Category	Features & Business Logic
Flight Volume (2017 full year)	total_flights_2017, total_distance_2017, months_active_2017, total_points_acc/red_2017, total_dollar_redemption. Captures full-year scale of engagement. A member active in 10/12 months is structurally different from one in 2/12.
Recency & Momentum	flights/points_last_6m (H2 2017 window), months_active_last_6m, flight_trend_h1_to_h2, recency. Volume misses' direction a member who flew heavily in Jan but not since Sep is far more at risk than one flying consistently all year.
Points Economics	points_balance, points_balance_pct, redemption_rate, points_per_flight. High balance + declining flights = loyalty trap (accumulated value keeps member enrolled but not engaged). High redemption = program delivering real value.
Tenure & Lifecycle	tenure_months, years_enrolled, is_new_member. Longer-tenured members churn less. New members (2017 enrollees) have no historical flight data and behave structurally differently.
Seasonal Decomposition	flights_autumn/spring/summer/winter + share_* columns. Canadian travel patterns vary substantially by season; unusual concentration in one season is a signal.
Interaction Terms	tier_x_recent_silence (Aurora silent > Star silent), tenure_x_silence (long-tenured going quiet is alarming), clv_x_negative_trend (high-value member with declining flights = worst churn scenario).

2.2 Model: Random Forest with Class Weighting

Algorithm chosen for: (1) interpretable feature importances for business stakeholders; (2) robustness to mixed feature types without normalization; (3) class_weight="balanced" compensates for the 7.8% minority class — without it, predicting everyone as retained gives 92% accuracy and catches zero churners.

Validation: 5-fold Stratified KFold, ensuring each fold maintains the 7.8% churn ratio. All churn probabilities are Out-of-Fold (OOF) — every member scored by a model that never saw them. Honest, not training-fit.

Model Performance

Fold 1: 0.844 · Fold 2: 0.838 · Fold 3: 0.840 · Fold 4: 0.843 · Fold 5: 0.836
Mean OOF AUC: 0.840 ± 0.003

At the operating threshold (90th percentile): ~6–7 of every 10 High-Risk members are genuine churners. Versus the 7.8% base rate, this is an ~8× improvement over random targeting.

Leakage checks: (1) Single-feature AUC no feature exceeded 0.75 alone. (2) 2018 columns excluded from feature matrix. Model is genuinely predicting.

2.3 Risk Tier Assignment

Quantile cuts on OOF probability distribution: Low Risk = bottom 70% (11,757 members) · Medium Risk = 70th–90th percentile (3,378) · High Risk = top 10% (1,602 members). Churn rates rise monotonically across tiers model discrimination confirmed.

3. CUSTOMER SEGMENTATION & VALUE ANALYSIS

3.1 Why CLV Alone Fails

CLV is a lagging indicator. A member with £27,000 CLV who flew last month and one with £27,000 CLV who hasn't flown in 6 months have radically different forward value. The segmentation separates these groups and makes that difference operationally visible.

3.2 Method: K Means on RFM + Redemption Rate

K=4 clusters on standardized features: Recency (months since last 2017 flight), Frequency (total 2017 flights), Monetary (CLV), Redemption Rate. Clusters named by empirical profile rules not by arbitrary integer ID ensuring stability across re-runs.

Segment	Members	Avg CLV	Avg Flights (2017)	Churn Rate	High-Risk Count
Premium Loyalists	1,109	£27,729	15.4	7.7%	82
Frequent Flyers	9,755	£6,380	18.9	6.0%	662
Active Engagers	1,564	£6,638	13.5	9.2%	230
Dormant Members	4,309	£7,041	0.2	11.5%	628

Premium Loyalists: 4.4× average CLV. Fly regularly, redeem actively. Losing one costs more than losing five average members. Proactive management required even at Low Risk not just rescue at High Risk.

Frequent Flyers: Largest segment (58% of members), highest absolute churn count (584), but lowest churn rate. Low redemption rate suggests habit-driven loyalty more durable but less responsive to points offers.

Active Engagers: Highest redemption rate among non-dormant segments. Loyalty is rewards-driven, not habitual. When program value drops, they leave faster than Frequent Flyers.

Dormant Members: 0.2 avg flights in 2017 with above-average CLV (£7,041). These are not never-actives they were valuable and stopped. Highest churn rate (11.5%) and largest High-Risk pool (628). Most immediately actionable segment.

Three findings CLV alone would have buried:

- Dormant Members look 'medium value' on CLV but are the highest-churn-risk group the behavioral lens reveals the danger.
- Program's 18% growth in active members masked 1,310 departures. New members offset losses; headline hid the problem.
- Redemption rate predicts churn better than flight frequency alone. Economically rational members (Active Engagers) leave when value drops; habit-driven members (Frequent Flyers) stay longer.

4. SMART RETENTION — THE 4×3 ACTION MATRIX

Every cell specifies: who receives the action, when, what offer, which channel, and why. No vague recommendations. Actions are calibrated: higher risk = more intensive and expensive intervention.

Segment · Risk	Action	Offer	Rationale	Timing	Channel
PREMIUM LOYALISTS (1,109 members · avg CLV £27,729 · High-Risk: 82)					
High Risk	Loyalty call	Complimentary upgrade + 2× points for 60 days	Losing one high-CLV member costs more than any intervention. Phone signals individual recognition.	Within 7 days	Phone loyalty team
Medium Risk	Re-engagement email	1.5× points on next 3 flights	Early warning — act before risk escalates.	Within 14 days	Personalized email
Low Risk	VIP reward	Quarterly surprise bonus points	Proactive maintenance is cheaper than rescue at high risk.	Each quarter	Email + app
ACTIVE ENGAGERS (1,564 members · avg CLV £6,638 · High-Risk: 230)					
High Risk	Redemption campaign	Double value on next points redemption	Redemption is their hook target the mechanism that keeps them engaged.	Within 14 days	Email + app push
Medium Risk	Check-in email	1.5× points on next flight	Recently active small nudge is sufficient.	Within 21 days	Email
Low Risk	Newsletter	Program updates + promotions	Healthy segment maintain with near-zero cost.	Monthly	Email
FREQUENT FLYERS (9,755 members · avg CLV £6,380 · High-Risk: 662)					
High Risk	Tier fast-track	Next tier if 3 flights in 60 days	Frequent Flyers respond to status. A tier challenge locks in future behavior.	Within 14 days	Email + app push
Medium Risk	Points expiry alert	Balance reminder + redemption options	Low redemption rate nudging redemption re-engages.	60 days before expiry	Email
Low Risk	No action	—	Low-risk frequent flyers save budget for higher-risk segments.	Monitor quarterly	—
DORMANT MEMBERS (4,309 members · avg CLV £7,041 · High-Risk: 628)					
High Risk	Last-chance win-back	50% discount on flight booked within 14 days	Already inactive and high-risk — aggressive offer or accept loss. SMS added: member may have disengaged from email.	Immediately	Email + SMS

Medium Risk	Reactivation nudge	500 bonus points for logging in + browsing	Test cheap re-engagement before committing larger budget.	Within 30 days	Email
Low Risk	No action	—	Dormant but low-risk seasonal or infrequent flyers, not worth spend.	—	—

5. BUSINESS RECOMMENDATIONS & LIMITATIONS

Three Recommendations for the CFO and CMO

Rec 1: Launch a Silent High-Value Rescue Protocol

Target: 130 silent disengages (Signal 2) + 628 Dormant High-Risk members · Urgency: Immediate
 These members have above-median CLV but zero 2017 flights and no formal cancellation — they were valuable and stopped without leaving a trace. Deploy the last-chance win-back (50% flight discount, 14-day window, email + SMS). CFO framing: retaining 10% of 628 dormant High-Risk members at £7,041 average CLV = £4.4M value protection, against intervention cost well below that.

Rec 2: Protect Premium Loyalists Before They Signal Risk

Target: All 1,109 Premium Loyalists · Urgency: Ongoing
 6.6% of members, 4.4× average CLV. Do not wait for churn signals. The 82 High-Risk Premium Loyalists get a personal loyalty ambassador call immediately. The 771 Low-Risk ones get quarterly VIP rewards regardless proactive maintenance is cheaper than rescue. CMO framing: airlines that win in loyalty make their highest-value members feel individually known, not batch-emailed.

Rec 3: Redirect 20% of Acquisition Budget to Targeted Retention

Target: Program-level budget allocation · Urgency: Strategic
 Beneath 18% membership growth, 1,310 members churned ~£11M in departing CLV (£8,394 avg per churmer). Reducing churn from 7.8% to 5% preserves ~470 members/year = £3.9M annually. Every member in the database already has an OOF churn probability and risk tier; intervention priority lists are directly exportable.

Limitations & Honest Trade-offs

Limitation	What it means in practice
No pre-2017 flight data	Activity covers 2017–2018 only. Signal 2 uses CLV + enrollment year as proxies for prior engagement reasonable but not directly observed. A dataset from 2012 onwards would strengthen the behavioral churn signal.
Uncalibrated probabilities	OOF probabilities are well-ordered but not calibrated: 0.30 ≠ 30% chance of churn. For precise budget optimization (expected value per intervention), add Platt scaling or isotonic regression calibration.
Analytical churn definition	Our 3-signal definition may differ from the airline's operational definition (e.g., only formal cancellations). Before production deployment, align the label with the marketing and ops team.
KMeans cluster stability	Cluster IDs shift between runs without random_state=42. The naming rules (highest CLV → Premium Loyalists, etc.) must be re-applied on each refresh to maintain stable segment labels.

CONCLUSION

This analysis produced three deployable outputs: a churn model (OOF AUC 0.840), a behavioral segmentation into four actionable clusters, and a 12-cell retention matrix specifying exact interventions for every segment-risk combination. The central finding is structural CLV alone does not capture churn risk. A member silent for six months with high CLV is in more danger than one with the same CLV who flew last week. Behavioral signals are the true leading indicators of departure.

The 2018 growth headline masked a real problem: 1,310 departing members, £11M in leaving CLV. The framework built here allows the marketing team to identify the 1,602 highest-risk members, deploy calibrated interventions, and measure conversion ROI by segment and tier. The data and the action lists are ready.

Analytical Deliverables

- ▶ **Churn model: Random Forest, 5-fold OOF AUC 0.840, leakage-free**
- ▶ **16,737 members scored with honest churn probabilities**
- ▶ **1,602 High-Risk members identified for immediate action**
- ▶ **4 behavioral segments : meaningfully distinct and operationally actionable**
- ▶ **12-cell action matrix : action / offer / timing / channel / rationale for every cell**
- ▶ **Exports: retention_actions_all.csv · retention_actions_urgent.csv**
- ▶ **3 business recommendations with financial framing**